BJT operation and DC bias I

Practice questions · Day 06 · Thu 29 Oct · pack E4

5. Practice

From your course files

WS4 Q6 (course question).
6.1. Which one of the following statements is correct for a BJT transistor?
A) BJT collector current can be controlled by the voltage between The Base and emitter
B) In a PNP transistor, the Emitter is an n-type semiconductor
C) A BJT has three PN junctions that form the three terminals
D) In a PNP transistor, the Collector is an n-type semiconductor
6.2. In a PNP BJT transistor,
A) the base-emitter junction is reverse biased in the cut-off mode
B) the collector-base junction is forward biased in the cut-off mode
C) the emitter-base junction is reverse biased in the cut-off mode
D) the emitter-collector junction is reverse biased in the cut-off mode
6.3. In an NPN BJT in the active mode of operation, the Base-Emitter PN junction is forward-biased so that
A) electrons flow from base to emitter
B) holes flow from emitter to base
C) electrons flow directly from emitter to collector
D) holes flow from emitter to base

WS4 Q7 (course question). For a BJT transistor, find the values of β for the values of α = {0.5, 0.8, 0.98, 0.999}.

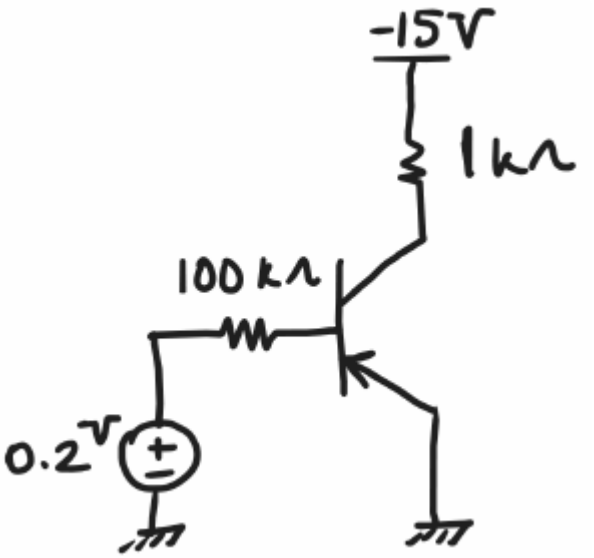
WS4 Q8 (course question). Consider a PNP transistor in active mode with a current gain of β = 50. Find,
1. Base (I_B) and emitter (I_E) currents when collector current is $I_C = 4$ mA.

2. Collector (I_C) and base (I_B) currents when emitter current is $I_E = 4$ mA.
3. Collector (I_C) and emitter (I_E) currents when base current is $I_B = 100$ µA.

WS4 Q9 (course question). Consider a transistor in the active mode region. Find,
1. Emitter current (I_E), β and α when collector and base currents are $I_C = 8$ mA and $I_B = 100$ µA.
2. Collector current (I_C), β and α when emitter and base currents are $I_E = 5$ mA and $I_B = 100$ µA.
3. Base current (I_B), β and α when collector and emitter currents are $I_C = 10$ mA and $I_E = 10.1$ mA.

WS4 Q10 (course question). Consider a transistor with a gain β = 50. For each of the following cases identify the operating mode (active, cut-off and saturation) of the transistor.
1. $I_B = 0$ µA and $I_C = 0$ mA.
2. $I_B = 100$ µA and $I_C = 5$ mA.
3. $I_B = 100$ µA and $I_C = 1$ mA.
4. $I_B = 200$ µA and $I_C = 10$ mA.

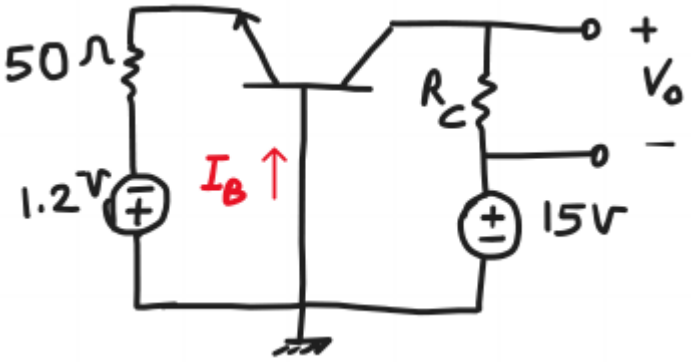
WS4 Q11 (course question). In the circuit below, replace the PNP transistor with its large-signal equivalent model if the transistor is in,
1. Active mode.
2. Saturation mode.
3. Cut-off mode.



WS4 Q11 circuit (from the workshop sheet)

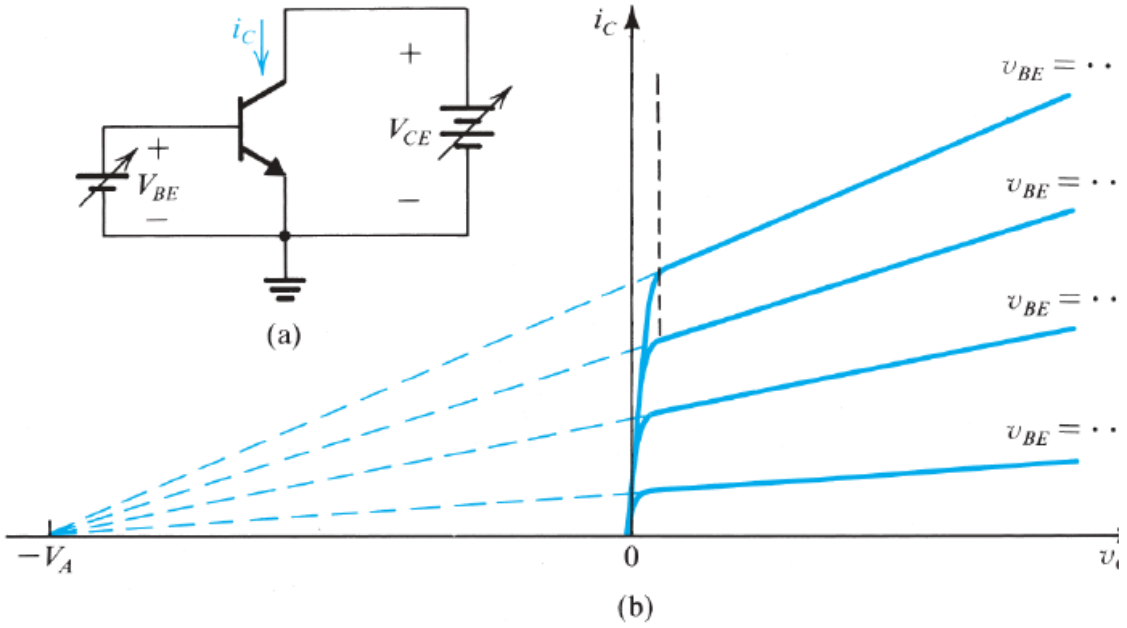
WS4 Q12 (course question). In the circuit below, replace the NPN transistor with its large-signal equivalent model if the transistor is in,

1. Active mode.
2. Saturation mode.
3. Cut-off mode.



WS4 Q12 circuit (from the workshop sheet)

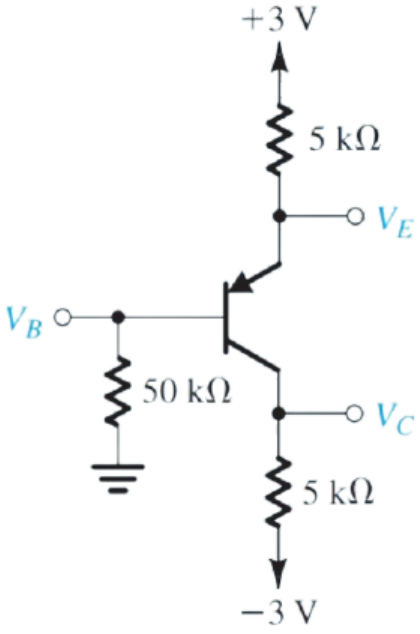
WS5 Q1 (course question). A practical $i_c - v_{CB}$ characteristics of a BJT are shown in the figure below. Identify active, saturation, cut-off regions, and early voltage. Also, explain what the non-zero slope means and how you expect the v_{BE} value to change from lower i_C to higher i_C.



WS5 Q1 figure (from the workshop sheet)

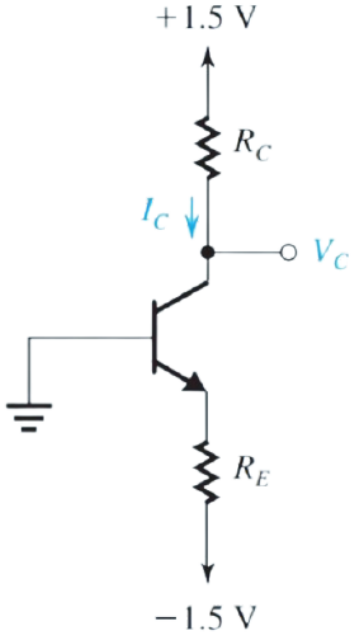
WS5 Q3 (course question). A single measurement indicates the emitter voltage of the transistor in the circuit below to be $V_E = 1$ V. Under the assumption that $V_{EB} = 0.7$ V and transistor is in active mode, find:

1. Emitter, collector and base currents.
2. Base and collector voltages.
3. α and β.



WS5 Q3 circuit (from the workshop sheet)

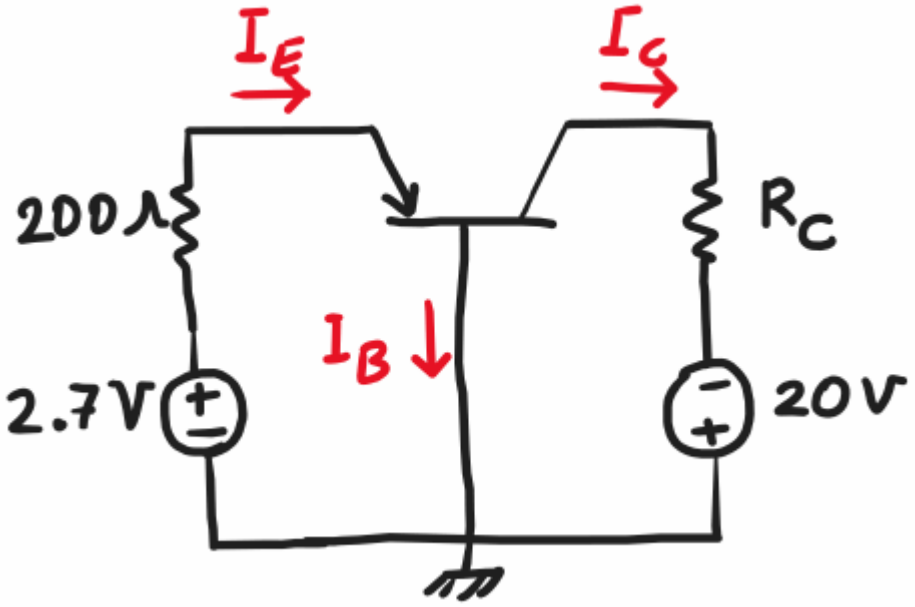
WS5 Q4 (course question). In the circuit below, the transistor is in active mode and has $V_{BE} = 0.7$ V and β = 100. Select resistors R_C and R_E such that $I_C = 0.2$ mA and $V_C = 0.5$ V.



WS5 Q4 circuit (from the workshop sheet)

WS5 Q5 (course question). In the circuit below, the PNP transistor has α = 0.9 and $V_{EB} = 0.7$ V (sometimes, we say $|V_{BE}|$ to avoid PNP and NPN voltage polarity confusion). Assume that the transistor is in active mode. Find,

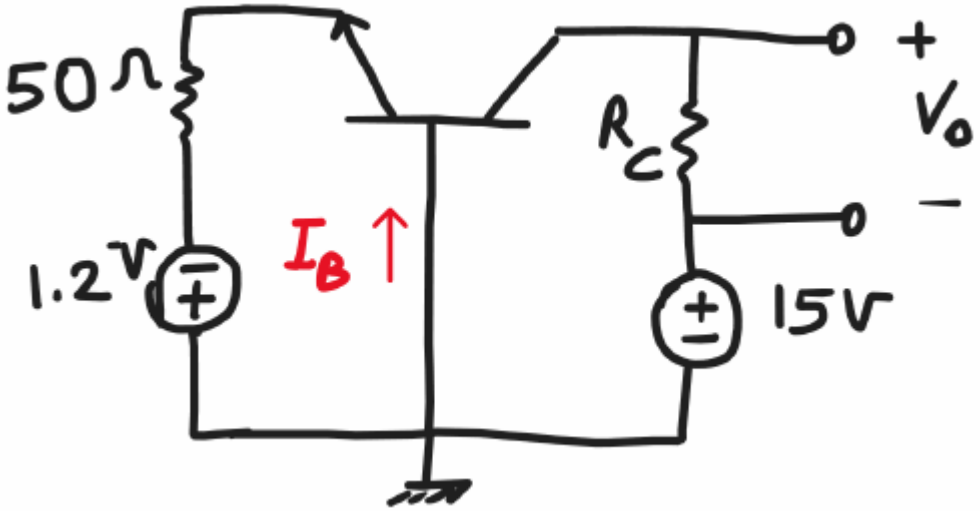
1. Emitter, collector and base currents.
2. What should be the minimum value of R_C if $|V_{CE,max}| = 20$ V?



WS5 Q5 circuit (from the workshop sheet)

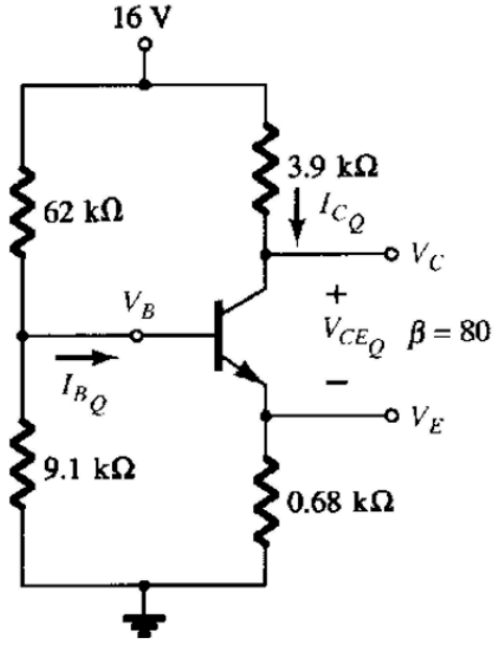
WS5 Q6 (course question). In the circuit below, the NPN transistor has α = 0.9 and $|V_{BE}| = 0.7$ V. Assume that the transistor is in active mode. Find,

1. the value of R_C if we want $|V_o| = 4.5$ V.
2. V_{CE} based on the value of R_C obtained previously.



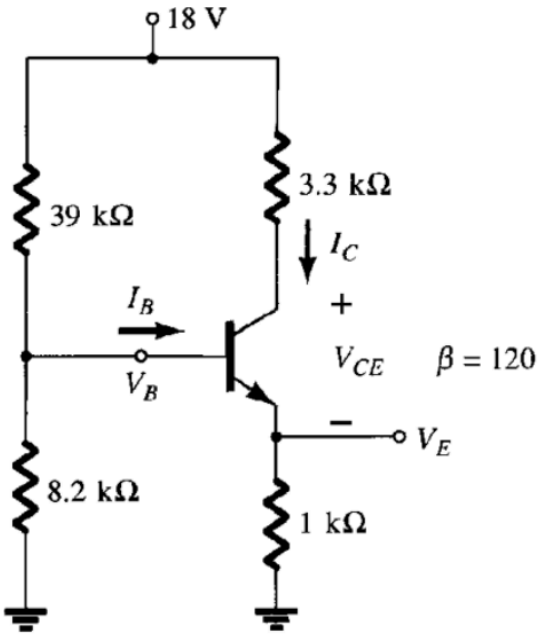
WS5 Q6 circuit (from the workshop sheet)

WS5 Q7 (course question). For the voltage-divider configuration in the circuit below, determine (a) I_{BQ}, (b) I_{CQ}, (c) V_{CEQ}, (d) V_C, (e) V_E and (f) V_B.



WS5 Q7 circuit (from the workshop sheet)

WS5 Q8 (course question). Determine the following for the voltage-divider configuration in the circuit below, using the approximate approach if the required condition is satisfied. (a) I_C, (b) V_{CE}, (c) I_B, (d) V_E and (e) V_B.



WS5 Q8 circuit (from the workshop sheet)

WS5 Q14 (course question). In the circuit below, assume that the BJT has β = 50, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Find the base and collector currents and check the transistor operating mode.

WS5 Q14 circuit (from the workshop sheet)

PT Q6 (practice test, 6 marks). Draw the load line for the network below on the characteristics.

a. Find the Q-point and resulting values I_{C_Q} and V_{CE_Q}.

b. Find the value of β at the Q-point.

PT Q6 circuit (from the practice-test paper)

PT Q6 output characteristics (from the practice-test paper)

PT Q7 (practice test, 10 marks). For the circuit shown on the right, β = 50, $V_{BE,on} = 0.7$ V, and $V_{CE,sat} = 0.2$ V. Answer the following questions assuming that the BJT operates in the **active state**.

a) Find the Thevenin equivalent circuit (i.e., V_{Th} and R_{Th}) seen from the base of the transistor. [4 marks]

b) Redraw the circuit by replacing the transistor with its **DC equivalent circuit** model and Thevenin equivalent model in the base. [4 marks]

c) Determine collector current (I_C) and voltage (V_C). [2 marks]

PT Q7 circuit (from the practice-test paper)

New questions

New question 1 [4 marks, easy]. A transistor has β = 99 and $I_{CBO} = 10$ nA.

a. Find α. [1]
b. Find I_{CEO}. [1]
c. With $I_B = 20$ μA, find I_C and I_E including leakage, and comment on whether leakage matters here. [2]

New question 2 [5 marks, easy]. Fixed bias: $V_{CC} = 15$ V, $R_B = 470$ kΩ, $R_C = 3.3$ kΩ, emitter grounded, β = 100, $V_{BE} = 0.7$ V.

a. Find I_{BQ}, I_{CQ} and V_{CEQ}. [3]
b. Give the load-line intercepts and confirm the mode using V_{BC}. [2]

Circuit for New question 2: fixed bias.

New question 3 [6 marks, medium]. Emitter-stabilised bias: $V_{CC} = 18$ V, $R_B = 510$ kΩ, $R_C = 2.4$ kΩ, $R_E = 1.5$ kΩ, β = 100.

a. Find I_B, I_C and V_{CE}. [3]
b. Find V_E, V_B, V_C and V_{BC}. [2]
c. Find $I_{C,sat}$. [1]

Circuit for New question 3: emitter-stabilised bias.

New question 4 [8 marks, medium]. Voltage divider: $V_{CC} = 15$ V, $R_1 = 47$ kΩ, $R_2 = 10$ kΩ, $R_C = 2.7$ kΩ, $R_E = 1.2$ kΩ, β = 150.

a. Test whether the approximate method is valid. [1]
b. Find I_C, V_{CE} and I_B by the approximate method. [3]
c. Repeat by the exact method and give the % error in I_C of the approximation. [4]

Circuit for New question 4: voltage-divider bias.

New question 5 [10 marks, hard, chained]. Voltage divider: $V_{CC} = 12$ V, $R_1 = 33$ kΩ, $R_2 = 12$ kΩ, $R_C = 2.2$ kΩ, $R_E = 0.56$ kΩ, β = 60.

a. Test the approximation. [1]
b. A student ignores the test and uses the approximate method anyway. What I_C and V_{CE} do they get, and what is physically wrong with the result? [3]
c. Do the exact analysis: I_B, I_C, V_{CE}, V_E, V_B, V_C. Confirm the mode. [4]
d. Find $I_{C,sat}$ and state where the Q-point sits on the load line. [2]

Circuit for New question 5: voltage-divider bias.

New question 6 [6 marks, hard, reading characteristics and design]. On a set of CE output characteristics, a load line runs from $V_{CE} = 24$ V to $I_C = 8$ mA. It crosses the $I_B = 30$ μA curve at $I_C = 4.2$ mA. The circuit is emitter-stabilised with $R_C = 2.4$ kΩ.

a. Find V_{CC}, $R_C + R_E$ and R_E. [2]
b. Find V_{CEQ} and β_{dc} at the Q-point. [2]
c. Find R_B, choose the nearest standard value (E12), and find the resulting I_B. [2]

Circuit for New question 6: emitter-stabilised bias, V_CC, R_E and R_B to be found from the load line.

6. Quick recall check

1. Convert α = 0.995 to β, and β = 150 to α.
2. An NPN has $V_B = 2.7$ V, $V_E = 2.0$ V and $V_C = 2.2$ V. Which mode is it in?
3. Write the base-loop equation for I_B in emitter-stabilised bias.
4. What condition allows the approximate voltage-divider method, and why?
5. What are the two intercepts of the DC load line for a circuit with R_C and R_E?